



Zhou Ziyi

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Education

Guangzhou University, School of Economics and Statistics	B.Sc. in Statistics	2022.09 - 2026.06
GPA: 3.65/4.00; Average Score: 90.16; Rank: 6/76 Major Courses: Probability Theory, Regression Analysis, Time Series, Data Mining, Statistical Computing, Stochastic Processes Honors & Awards: First-Class Comprehensive Scholarship (2025, 2026); Second-Class Comprehensive Scholarship (2023, 2024); Outstanding Student (2025, 2026); Outstanding Student Cadre; among 20+ university-level honors.		

Selected Research & AI-Enabled Data Projects

HKUST(GZ)	Research Assistant	2026.04 - Present
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Geospatial Data Analysis, Deep Learning, and Research Visualization

- Processed geospatial and social datasets to support AI-assisted spatial pattern discovery, research visualization, and interdisciplinary analysis.
- Organized preprocessing, experiment-tracking, result-comparison, and visualization materials for deep learning-based research workflows.
- Produced literature summaries, data summaries, visualization outputs, and report drafts to improve the reproducibility and interpretability of research materials.
- Converted model-generated spatial patterns into interpretable visual evidence, connecting **AI-assisted analysis** with human problem framing and contextual judgment.
- Documented selected **Python / QGIS / geospatial workflows** through **CSDN and GitHub**, turning research tasks into reusable technical notes and step-by-step procedures.

Data Pi THU, School of Software Engineering, Tsinghua University	Project Team Member	2025.03 - Present
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AI Technical Communication and Model Explanation

- Authored and translated technical articles** on Transformer architecture, reinforcement learning, AI agents, and multi-armed bandit problems.
- Produced structured AI explainers, including Layer-by-Layer Visual Analysis of Transformers and Essential Reading Before Reinforcement Learning.
- Converted complex algorithmic logic into visual, reader-friendly Chinese technical narratives for broader technical audiences.
- Improved public-facing AI communication by making abstract model mechanisms easier to understand, evaluate, and discuss.

Foshan Zhumiao Museum, Planning & Promotion Department	Data Analysis Assistant	2024.11 - 2025.04
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Human-Centered Cultural Data Science

- Analyzed **800+ survey responses** to evaluate public perception, audience engagement, and participation patterns related to Foshan lion dance and Intangible Cultural Heritage.
- Built a **Python / R-based** survey analysis pipeline covering **data cleaning, preprocessing, descriptive statistics, visualization, clustering, logistic regression, and factor analysis.**
- Identified audience segments and key drivers of cultural participation, providing evidence-based suggestions for museum branding, public engagement, and targeted communication.
- Converted survey findings into actionable communication strategies, supporting report writing, visualization, youth-oriented study-tour course design, and cultural IP development.
- This work contributed to award-winning outcomes, including **National First Prize in the Zhengda Cup Market Survey and Analysis Competition** and **Second Prize in the National Final of the National College Business Elite Challenge.**

Production Decision Optimization with GA-SA and Dynamic Programming	2024
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Team Leader | Optimization Modeling, Algorithm Design, and Decision Analysis

- Led a mathematical modeling project on inspection, disassembly, and rework decisions under defect-rate, cost, and profit constraints.
- Built a **multi-stage decision framework** integrating sampling inspection, genetic simulated annealing, dynamic programming, and cost-profit analysis.
- Implemented core optimization algorithms in **Python and MATLAB** to generate scenario-specific inspection and disassembly strategies.
- Reduced the average sampling workload by **19.4%** through an optimized sampling inspection plan while maintaining decision reliability.
- Translated numerical optimization results into interpretable decision rules for cost control, quality improvement, and profit-oriented production planning.

Applied Data Science Experience

Zhongyan Technology Co., Ltd.	Data Analysis Assistant	2025.09
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- Contributed to **SPSSPRO instructional materials** on multivariate statistical analysis, covering regression, discriminant analysis, clustering, PCA, factor analysis, and association rules.
- Converted statistical concepts, software operation logic, and applied examples into user-oriented method documentation.

- Produced clearer workflow explanations to help applied users understand model selection, interpretation, and implementation procedures.

Cainiao Network	Operations Trainee	2026.01 - 2026.03
• Analyzed cross-border e-commerce logistics workflows and digital operation scenarios within the Alibaba ecosystem. • Used Excel for data cleaning, visualization, and basic business analysis of user behavior and operational processes. • Proposed scenario-based optimization suggestions by combining user personas, business pain points, and workflow analysis.		
NetEase Media	Campus Content Operation Intern	2026.03 - 2026.05
• Planned and published 20+ original feature stories, supporting campus content operation and community engagement. • Managed “NetEase Little Bee” ranking-list operations, organizing 210+ candidate profiles and supporting 350+ interactive comments. • Reviewed content performance and user preferences to optimize publishing rhythm and improve audience engagement.		

Selected Awards

Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM)	Honorable Mention	May 2025
National College Student Market Survey and Analysis Competition	First Prize	May 2025
National College Business Elite Challenge: Cultural Tourism and Exhibition Innovation & Entrepreneurship Practice Competition	Second Prize	November 2025
National College Students Mathematical Modeling Competition	Second Prize, Guangdong Region	December 2024
National College Students Statistical Modeling Competition	Second Prize, Guangdong Region	July 2024
Lanqiao Cup National Software and Information Technology Talent Competition	Third Prize, Guangdong Region	April 2024

Campus Experience

Guangzhou University	School of Economics and Statistics Youth Association Publicity Department
Member and Deputy Director, Publicity Department	2022.10 - 2024.07
• Led publicity planning and new media operations for volunteer activities. • Managed WeChat posts, posters, videos, and event announcements, supporting online-offline public welfare communication. • Drafted and reviewed 20+ activity posts and helped improve student participation and institutional visibility.	

Technical Skills

- **Programming & Data Analysis:** Python, R, MATLAB, SQL, Excel, Pandas, NumPy
- **Statistical & ML Modeling:** Regression, clustering, factor analysis, time series, hypothesis testing, PCA, K-Means, Apriori
- **Optimization & Geospatial:** Genetic algorithm, simulated annealing, dynamic programming, nonlinear programming, QGIS, spatial visualization
- **Research & AI Workflow:** GitHub, LaTeX, PyCharm, VS Code, RStudio, Dify, Coze, AI technical writing

PROFILE & RBCC Fit

Statistics undergraduate with experience in AI-assisted geospatial analysis, human-centered cultural data science, and optimization modeling. My work focuses on how statistical models and AI workflows can support pattern discovery, research communication, and decision-making while human researchers provide context, validation, and social meaning.